1 Why probability?

- Most real world events have uncertain (or random) outcomes
  - flipping a coin
  - Tossing dices
  - Football games
  - Blackjack
  - Expected lifetime of iPhone
  - The actual resistance of a 100 Ohm resistor
  - ...
- How do we characterize these random events?
- Can we predict the outcome of a random event?

Probability Theory

- provides a complete set of mathematical tools and theories that can accurately and precisely describe the statistical behaviors of the random phenomenon.
- It is a branch of Mathematics
- It has a wide rang of applications to Engineers.

2 Applications

Random input signals

- E.g. the number of vehicles passing through a certain check points as an input to a traffic control system (e.g. duration of traffic lights)
- E.g. temperature fluctuations as an input to a heating system
- E.g. Music signals as an input to your stereo system

System reliability

- What is the expected life cycle of a given system?
- What is the probability of failure of a system?
  - Warranty duration, insurance policy, etc.

Computer simulation

- A cost effective and efficient way to test the performance of a system
- Random inputs, random system parameters (e.g. random component value, random disturbance, ...)

3 Set Theory

Set: a collection of elements

Example: $E = \{\alpha_1, \alpha_2, \ldots, \alpha_n\}$

- E is a set
- α_m is an element of the set E : $\alpha_2 \in E$

A **subset** of E is any set all of whose elements are also elements of E



Example: $E = \{1, 2, 3, 4, 5, 6\}$

- $F = \{1, 3, 6\}$ is a subset of E, $F \subset E$
- $G = \{1\}$ is a subset of E, $G \subset E$
- Any set is a subset of itself, $E \subset E$

Empty set: $\emptyset$ { }

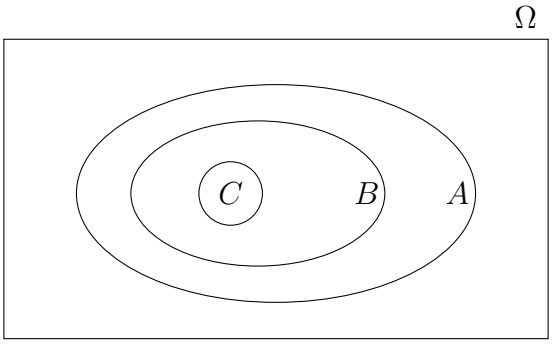
- A set that does not have any element
- Empty set is a subset of any set, $\emptyset \subset E$

Universal set Ω: the largest set of interests for a given application

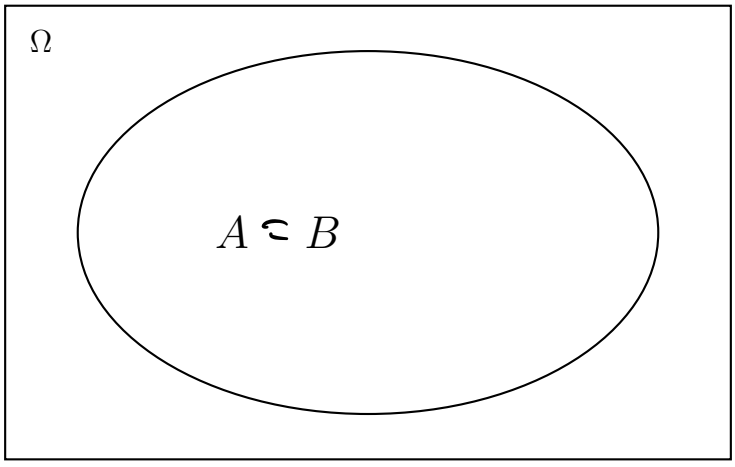
- E.g. toss a coin, $\Omega = \{H, T\}$
- E.g. roll a six-sided die, $\Omega = \{1, 2, 3, 4, 5, 6\}$

Venn diagram: A geometric representation, where the universal set Ω is represented by a square and the sets are represented by closed plane figures inside the square.

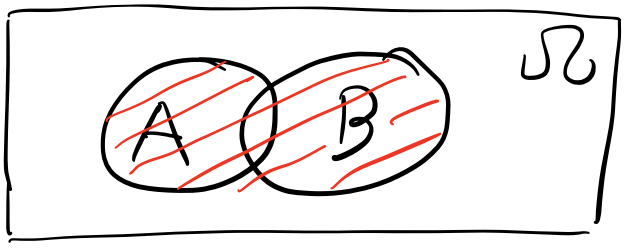
- E.g. $C \subset B \subset A$



- If $A \subset B$ and $B \subset A$, then $A = B$.



Union: The union of two sets A and B is a set consisting of all the elements from A or B or both, denoted as $A \cup B$.



$A \cup B$

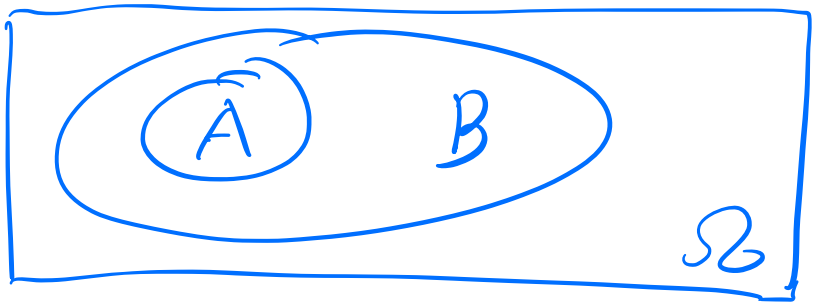
- Associate law: $A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C)$
- Commutative law: $A \cup B = B \cup A$
- Properties

$A \cup A =$ A

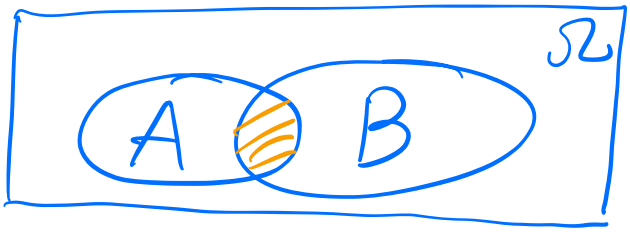
$A \cup \Omega =$ Ω

$A \cup \emptyset =$ A

- If $A \subset B$, $A \cup B =$ B



Intersection: The intersection of two sets is the set consisting of all the elements from both sets, denoted as $A \cap B$.



$A \cap B$

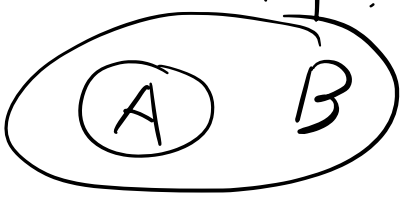
- Associative law: $A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$

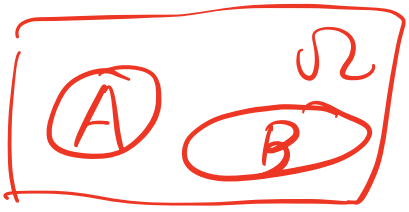
- Commutative law: $A \cap B = B \cap A$

- Properties

$$A \cap A = A$$
$$A \cap \Omega = A$$
$$A \cap \emptyset = \emptyset$$

- If $A \subset B$, $A \cap B =$ A

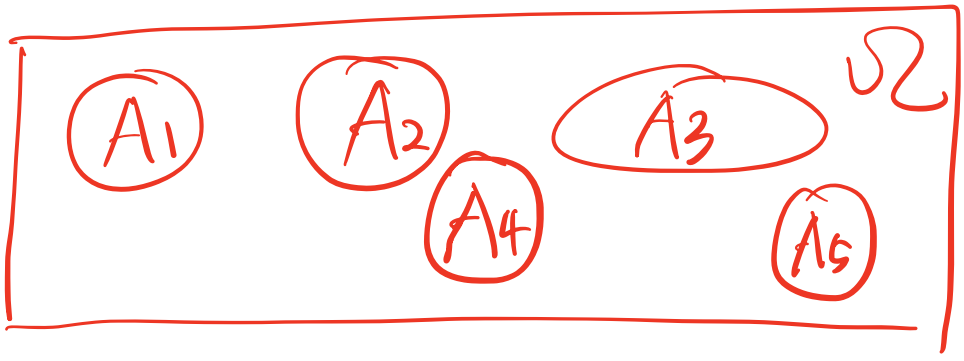




- Mutually exclusive (disjoint):
  - Two sets A and B are mutually exclusive if $A \cap B = \emptyset$.
  - A collection of sets $A_1, A_2, \cdots, A_n$ is mutually exclusive if and only if

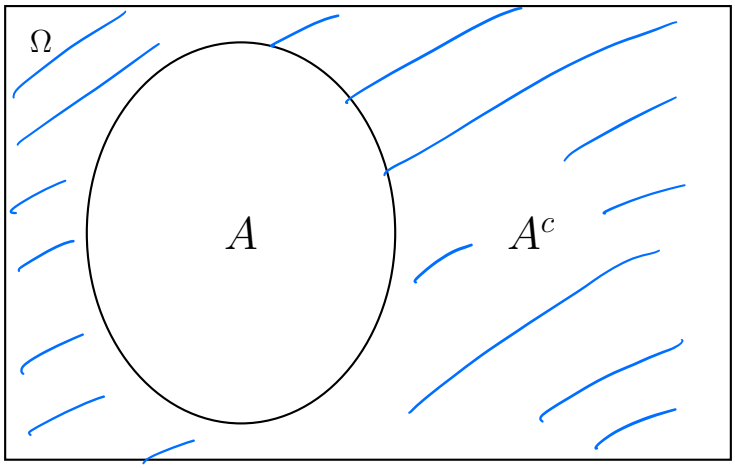
$$A_i \cap A_j = \emptyset, \quad \forall i \neq j.$$

for any



Complement: The complement of set A is a set containing all the elements of Ω that are not in A, denoted as A^c.

(or $\bar{A}$)



- Properties

$$\emptyset^c = \Omega \qquad \Omega^c = \emptyset$$
$$(A^c)^c = A$$
$$A \cap A^c = \emptyset \qquad A \cup A^c = \Omega$$

- If $A \subset B$, then $B^c \subset A^c$.
- DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

We need to show the following

(1) $(A \cup B)^c \subset A^c \cap B^c$

(2) $A^c \cap B^c \subset (A \cup B)^c$

To show (1):

$\forall\, x \in (A \cup B)^c$, we have $x \notin A \cup B$

$\Rightarrow x \notin A$ & $x \notin B$

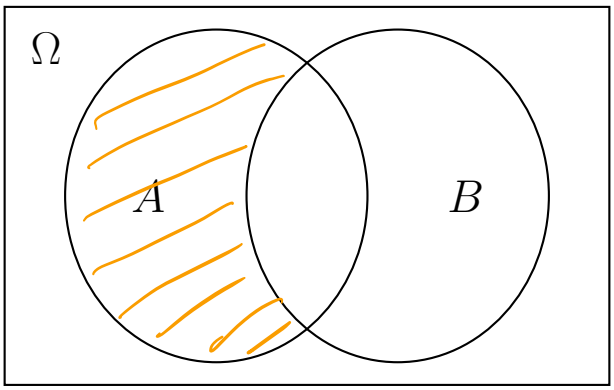
$\Rightarrow x \in A^c$ & $x \in B^c$

$\Rightarrow x \in A^c \cap B^c$

$\therefore (A \cup B)^c \subset A^c \cap B^c$

DIY to show (2).

Difference: The difference of two sets, $A - B$, is a set consisting of the elements of A that are not in B.



Examples

$A - \emptyset =$

$A - \Omega =$

$\Omega - A =$